Digital Twin Framework for Personalized Building Management in Ambient Assisted Living

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***Abstract*—Ambient Assisted Living (AAL) systems, which integrate digital healthcare with Smart Home concepts, have become increasingly important for supporting independent living, particularly for aging populations and individuals with physical impairments. This paper proposes a sensor-based Digital Twin (DT) framework that incorporates user-centric perspectives to enhance AAL systems and improve energy management in smart homes. To this end, the framework relies on a comprehensive ontology describing the semantic behaviour and relationships of relevant buildings, their occupants, and connected objects inside them. Unlike traditional approaches, which focus solely on building performance, this framework integrates occupants' preferences and habits as critical elements contributing to both energy efficiency and personalized care.**

Through data collection mechanisms and machine learning techniques, the framework provides occupants with valuable insights into how their activities, particularly those involving appliance usage, impact overall energy consumption in their homes. This is highly beneficial for AAL applications, enabling occupants to make informed decisions on sustainable and efficient energy use, and ensuring a safer and more comfortable living environment. As a result, the paper outlines the proposed ontology and the initial implementation of the system prototype, which leverages the Microsoft Azure platform and its Cosmo DB No-SQL database. The prototype is complemented with a user-centric awareness dashboard, constructed using Flask and accessible through a web application interface.

***Index Terms*—Digital Twin; ontology; Internet of Things (IoT); Ambient Assisted Living (AAL)**

I. Introduction

Smart Home automation has gained significant attention in recent years due to the rapid growth of the Internet of Things (IoT) [1]. Among the various applications of Smart Home technology, one of the most promising is Ambient Assisted Living (AAL), which integrates digital healthcare with Smart Home concepts to enhance the quality of life, particularly for aging populations and individuals with physical limitations [2]. AAL aims to create adaptive living environments that support independent living by using interconnected systems to sense, anticipate, adapt to, predict, and respond to human behaviours and needs [3]. A crucial aspect of this paradigm is the improvement of occupants' comfort and well-being [4]. To create an autonomous monitoring system capable of understanding the context and making decisions accordingly within a building, it is essential to gather information on how users interact with various devices and appliances in the home. This data enables the system to operate based on the habits and preferences of the users, thereby respecting their comfort levels and providing personalized assistance tailored to their specific needs.

In this context, adaptive scheduling of household appliances according to occupants' preferences enables people with physical impairments to improve their house-related energy efficiency easily. By integrating user behaviour patterns with energy management strategies, smart home systems can seamlessly optimize both comfort and efficiency, making the process virtually transparent to the occupant. Such a heterogeneous system is characterized by an intricate interplay among humans, objects, and buildings, whose complex dynamic interactions require a comprehensive and accurate modeling approach to be captured. In this context, the concept of Digital Twin (DT) has attracted increasing attention in the built environment domain [5]. DTs are commonly defined as the digital and virtual versions of physical entities and mimic their real-world behaviours. This virtualization enables the creation of simulation processes that utilize physical models, sensor updates, management history, and other data to integrate multi-discipline and multi-physical quantities [6].

This paper proposes a comprehensive sensor-driven DT framework designed to enhance energy management practices and refine AAL systems to better support the increasing demands of an older population in their everyday lives. In particular, by modeling how occupants' activities can affect energy performance, the system can achieve coherent and accurate appliance scheduling. The significance and novelty of the proposed framework compared to the existing literature, lie in introducing a user-centric perspective wherein occupants' preferences and habits are critical elements contributing to the building's energy performance. As a result, the output is a

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tool providing relevant information regarding the influence of occupants' activities and habits on building energy efficiency.

The paper is structured as follows: Section II examines the background on Digital Twins (DT) in building management. Section III describes the reference scenario and system. Section IV defines the ontology scheme for the proposed DT framework. Section V details the implementation of a preliminary prototype. Section VI concludes with key findings and future work directions.

II. Background

A. Digital Twin

The first introduction of the term DT was in 2003 when Grieves at the University of Michigan introduced the "digital equivalent to a physical product" definition [7]. It's possible to divide the DT into three main concepts: the physical twin, the virtual twin and the connection mechanism between the two components [8]. The physical twin is the real-world entity. The virtual twin is a representation of the virtual world, which mirrors the real-time behaviour of the physical twin. The connection mechanism is the relationship that permits the bidirectional data flow between the virtual and physical twins. The connection mechanism must be automatic and in real-time. In [9], there is an accurate definition of DT: "A DT is a living, intelligent and evolving model, being the virtual counterpart of a physical entity or process. It follows the life cycle of its physical twin to monitor, control, and optimize its processes and functions."

B. Digital Twins for Building Management

DTs are a tool of great importance for building management, simulation, and consumption forecasting. DTs enable design, testing, and optimization solutions before they are implemented in the real world, reducing costs and energy consumption, thus improving energy efficiency [10]. DTs are dynamic models that describe the behaviour of buildings over time. The DT technology is applied in building automation by generating virtual replicas of buildings. Based on it, building managers can explore various scenarios and assess their effects on energy usage. Furthermore, DTs can contribute to building energy management with useful energy consumption patterns and interfaces [11]. Overall, DTs can provide energy-saving solutions in diverse application fields such as smart cities, IoT implementations, industry 4.0, and smart manufacturing [12].

III. Proposed Methodology

A. Reference Scenario

The scenario revolves around a smart building equipped with diverse sensors, actuators, controllers, and smart devices. Its aim is to gather data concerning appliance energy consumption, renewable energy production, and occupants' daily activities and habits, facilitating informed decision-making processes. All the real entities that are part of the system and interact with it (i.e., person, device, appliance, building) have their correspondence in the virtual world, where they are represented by their DTs. DTs enable the system to visualize the information on energy consumption and production and, when relevant, the connection with occupants' activities that are useful for efficient energy management applications inside the building.

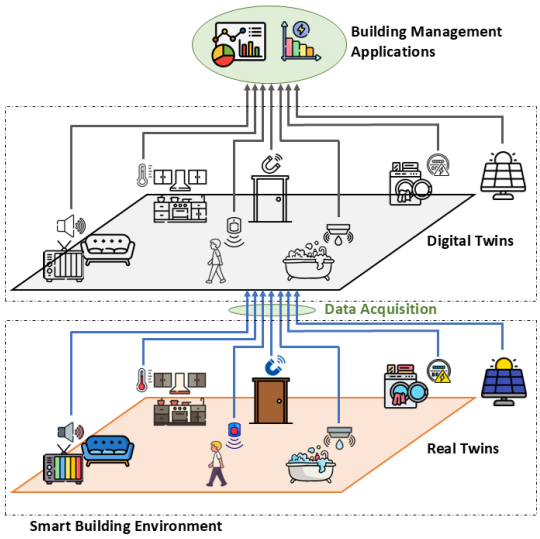


Fig. 1. Reference Scenario and Proposed System

Figure 1 shows an overview of this reference scenario. The lower layer of the image represents the physical world in which the Real Twins are present and operate. The data and information related to these entities are gathered and sent to their equivalent DTs. In the digital world, which is represented as the medium layer of the figure, this new information is used to change the state of the DTs accordingly, while all the old data are stored in a specific database. Finally, in the higher layer, Building Management Applications gather the DTs data and extract information metrics. This extracted information is shown to the final users in the form of energy consumption charts that help them understand the building's energy consumption/production trend. Indeed, thanks to this framework, the building occupants can have easy access to information about the overall energy consumption of the building, the consumption of each appliance in the house, the production of energy from RES, and information about their daily activities and habits and how they impact on the energy efficiency of the building. All the features of the Real Twins layer can be managed locally through smart devices, whereas those within the DTs and applications layer can be managed either locally or, more likely due to their high processing demands, in the Cloud.

B. Considered Requirements

To develop a DT-based system in the reference scenario, several key requirements must be addressed:

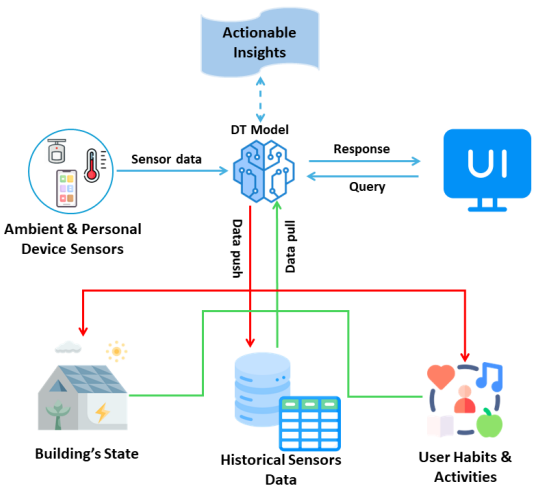


Fig. 2. Scheme of the System's Architecture

- *Context awareness*: it has to adapt its behaviour based on environmental conditions and occupants interactions.
- *Dynamism and adaptivity*: it has to be capable of responding to real-time changes within the building environment.
- *Mobility*: building occupants move outside of their reference building and from one building to another, keeping their personal objects with them. For this reason, the system must consider this mobility, while also ensuring that the virtualization of objects and occupants remains consistent even when they are outside the building premises.
- *Object relationships*: the virtualization of objects should allow for associations with either the building where they are located, or the person they are associated with or interact with, or both, enhancing the system's versatility and utility in various scenarios.
- *Occupants profiling*: The system needs to profile occupants, capturing their characteristics, behaviour, habits, and preferences.
- *Consumers and producers profiling*: appliances and RES have to be associated with their dynamic energy consumption and production profiles.

Addressing these key requirements is essential for the successful development and implementation of a robust DT-based system in the reference scenario. As it is more thoroughly explained in the following Sections, meeting such requirements ensures that the system can effectively model and simulate the complex interactions within buildings, objects and occupants, thus leading to a comprehensive knowledge of the context.

C. System Architecture

The System's Architecture considers the presence of different elements and modules, as shown in Figure 2. The data from ambient and personal devices' sensors are sent to the DT Model, which collects them. These new records are consistently stored in the Historical Sensors Data module of Figure 2, and the event time series is used to delineate different essential aspects within the system. Indeed, the recorded events themselves enable real-time monitoring of all sensors and device statuses while are also used to update information about users and buildings. Therefore, the model needs to store the new data and read historical records. Relying on sensor readings and records, the DT can, for example, identify the activity the user is performing at any time. If the user is showering, the DT Model should not only update the User Habits and Activities entity to reflect the ongoing activity but also adjust the Building's State element to account for potential changes in the home environment, such as increased temperature and humidity levels in the bathroom. Therefore, both the user's and the building's information need to be modified accordingly. Additional sensor data, like outdoor temperature and weather conditions, can further refine the Building's State. Furthermore, the DT also needs access to the Building's State information to conscientiously correlate this data with data obtained from the sensors and the User module. At the same time, access to User Habits and Activities information is necessary to confirm that the observed activity sequences are consistent with typical user behaviour. Finally, the DT module performs operations and aggregation functions on data to attribute them to appropriate context-aware relationships and intersect their effects. In this way, the system can provide end-users with user-centric tools, accessible through simple Interfaces, to enhance their awareness regarding energy efficiency within the building.

IV. Proposed Ontology

To follow the design choices outlined in Section III-B and the architecture's description in Section III-C, the creation of DTs was carried out through a conceptual subdivision of the three virtual entities, as it is evident from such requirements, buildings, objects and humans present distinct characteristics and behaviours. This choice led to the implementation of three different types of virtual model representations: Object DTs (ODTs), Personal DTs (PDTs), and Building DTs (BDTs).

The semantic modeling of these three entities, along with their characteristics and relationships, in the form of an ontology, is represented in Figure 3. A more detailed description of the proposed ontology is provided in this work [13]. The use of an ontology provides a structured framework that enables the definition and organization of the complex relationships and attributes of DT data. By formalizing data representation semantics, ontologies enable interoperability, semantic reasoning, and enhanced data integration across diverse applications and domains within the DT ecosystem.

A. Reference ontologies

One of the main available ontologies in building energy applications is the Smart Appliances REFerence (SAREF) Ontology [14]. SAREF focuses on tangible objects designed to accomplish a particular task. Each object is characterized

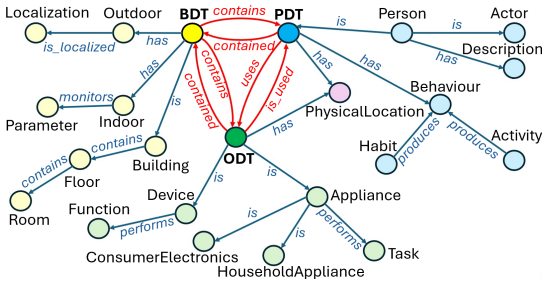


Fig. 3. Scheme of the main elements of the proposed ontology

by one or more functions, a description through properties, attributes, classes to which they belong, and information on energy consumption.

The Building Topology Ontology (BTO) [15], developed by W3C Linked Building Data Community Group, aims to provide a minimal ontology for describing relationships between a building's sub-components. It is designed to be extensible and models relationships within a building using Zones, Elements and Interfaces. The Semantic Sensor Network (SSN) [16] ontology, including its core Sensor, Observation, Sample, and Actuator (SOSA) [17] elements, is designed to help modelers describe sensors and observations, procedures, features of interest, samples, observed properties, and actuators. Other significant examples of building-related ontologies are RealEstateCore (REC) [18] and Brick Schema [19].

B. Object Digital Twin Ontology

The former class includes connected actuators, sensors, interfaces and controllers. The latter class describes the profile of household appliances, either smart or not, that are inside the building and, therefore, interact with the system. Examples of such appliances are fridges, washing machines, and heat pumps.

The relationship ODT-BDT is relevant from a wide-range point of view, as it involves processing and aggregating data from ODTs, ultimately fostering context awareness within the building. The ODT also interacts passively with the PDT, given that the functionality and consumption of the objects (both Devices and Appliances) contribute to occupants' profiling.

C. Personal Digital Twin Ontology

PDTs represent the human profile that interacts with the system. PDTs have active relationships with ODTs and passive relationships with BDTs, as illustrated in Figure 3. The PDT-ODT relationship is active as the ODTs provide data on occupants' activities, considering the direct or indirect actions they perform. Direct actions happen when occupants interact with a device to carry out a specific task (for example, using the washing machine to do the laundry). On the other hand, indirect actions entail the detection of occupant activities through sensors (for instance, PDTs are updated based on their presence in a certain room). Accordingly, PDTs perform actions that can modify the state of the building dynamic model.

The profiling of human behaviour is performed through the processing of the monitored parameters (such as object usage and sensing) and prediction of the actions of the occupants interacting with the system, as described more in detail in [20]. Here, the recognition and prediction of occupants' activities are performed by implementing a Machine Learning algorithm based on a Naive Bayes classifier, i.e., a type of classifier that applies Bayes' theorem to assume independence between features. Profiling takes place based on the history of actions (Habit) and its update based on the current detected action (Activity). The final output (Behaviour) occurs when the components of Habit and current Activity are added together.

D. Building Digital Twin Ontology

The BDT represents the highest level of the presented architecture, as it models the building behaviour. BDTs are composed of one or more floors, each including one or more rooms. Based on their main functionality, BDTs are categorized as residential or not residential. Residential buildings (e.g. houses, apartments, residences) describe domestic usage. Characteristic rooms are, e.g., bathroom, bedroom, and kitchen. Non-residential buildings (e.g. schools, factories, offices) model any other type of non-residential building.

As described in Figure 3, BDTs have both *internal* and *external* interactions. Internal interactions are those that the building has with the entities present inside it. The BDT, seen as a simulation of the building, is closely related to the functionality of the ODTs that are part of the system and is similarly related to its occupant activities.

Instead, external interactions are identified from a macroscopic perspective of the building. These interactions concern the relationships of the single BDT associated with a building with other BDTs, which consists of the mutual exchange of information. An example of this scenario is the case of an energy community, where the exchange of data regarding parameters for the correct coordination of optimized energy management is necessary. Another application case is when an occupant passes from one building to another. In this case, a PDT interacts sequentially with two BDTs, so an exchange of data is needed to ensure data consistency.

The BDT interacts with ODTs, which provide data that build the state profile of the building. In the same way, occupants affect the state of the system with their presence, so the PDTs contained inside the building collaborate to update the instances of the BDT.

Integrating the three digital entities allows for optimizing resources, enhancing user comfort, and minimizing consumption in AAL environments while offering flexibility and scalability to meet dynamic user needs.

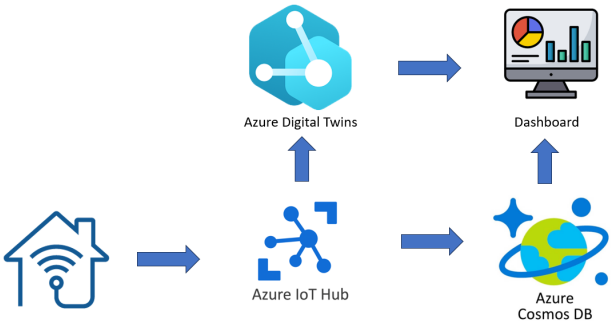


Fig. 4. First Prototype Functionality Blocks

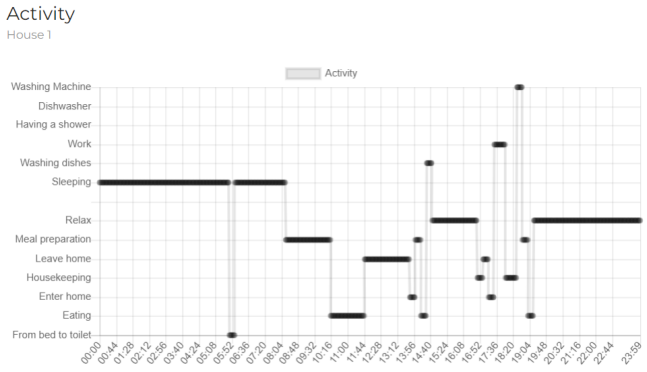


Fig. 5. Dashboard with the occupant activities over time (in min) in one day.

V. Implementation of the First Prototype

To validate the proposed ontology, a prototype was implemented through the Microsoft Azure platform. Azure is Microsoft's public cloud platform that offers a wide range of services. Azure was selected for this task because of the ad-hoc service for managing DTs.

To fully describe the behaviour and the interactions between buildings while maintaining a reasonable atomic point of view, this first prototype counts two different residential buildings.

To populate the system, three datasets were selected: CASAS [21], DEDDIAG [22] and Photovoltaic (PV) dataset from UK Power Networks [23].

CASAS is a Washington State University project that collects data from various sensors in homes to understand domestic activities and energy consumption patterns. The dataset is derived from on-site measurements using motion, contact, temperature, and light sensors. DEDDIAG is a dataset that focuses on the electricity consumption of individual household appliances. Recorded in 15 homes over up to 3.5 years, it includes power measurements from 50 devices such as dishwashers, washing machines, and refrigerators. The PV Solar Panel Energy dataset was developed by UK Power Networks to validate the impact of photovoltaic panels on energy production. Measurements were taken every hour, 10 minutes or minute.

According to these datasets, the first building is modeled as a single-family residential building with a coffee machine, a washing machine, a workstation, a refrigerator, and a dishwasher. There is a person who interacts with the system. More in detail, CASAS's dataset number 17 (Aruba) was examined because it reflects the requirements for using tags regarding occupants' behaviour and analyzing their activities. Therefore, activity recognition is developed through consumption data of the relevant household appliances used to carry out the activity and/or through data relating to the motion sensors placed in the areas of interest. The activities used to populate the PDT are: Bed to Toilet, Eating, Enter Home, Housekeeping, Leave Home, Meal Preparation, Relax, Sleeping, Washing Dishes, Working, Dishwasher utilization, and Washing Machine usage.

In this early stage, the use of household appliances was focused only on the coffee machine and washing machine. For this, the dataset related to House 8 of the DEDDIAG dataset was selected. The choice to analyze only the coffee machine and the washing machine as household appliances was made due to the fact that they represent the most relevant electric consumption in the scenario. Hence, the ODT has two instances populated according to the energy consumption of these two appliances.

The second building of this prototype is modeled as a single-family residential home with an electric heat pump and a PV system. For this reason, the PV dataset was used to describe the home's energy production. Data from DEDDIAG are used for the electric heat pump's consumption. House 14 was selected for this purpose. The resulting energy consumption profile is described by the BDT associated with this building.

The prototype consists of three functional blocks: data acquisition, simulation of DT working and graphical representation of the results. In Figure 4, the schema of the first prototype functionalities is reported. The updating phase of the DT instances takes place through the Azure IoT hub service, which allows the simulation of a sensor network workflow. Indeed, at this stage, the readings from real sensors are simulated by a local script in Python.

The IoT hub also facilitates real-time updates and the implementation of data history storage from the virtual sensor network through database connectivity. The storage functionality is carried out via the NoSQL Azure CosmosDB service.

In the Azure CosmosDB service, status changes and measurements are stored to create a historical record for the project's first prototype. The data saved in the database can concern ODT (e.g., consumption, environmental parameters), PDT (e.g., activity history, descriptive information), or BDT (e.g., descriptive data of the building, indoor and outdoor parameters). The database presents the "databaseCDB" instance, within which the "containerCDB" instance is saved, and the messages coming from the IoT hub are also saved. The messages are saved in JSON format and the "Body" contains

information regarding the source device. For example, considering a message related to the electric heat pump consumption, the fields of the "Body" would be the device ID, device name, measurement date and consumption value.

In this preliminary phase, data visualization was implemented via a Web App dashboard developed with Flask. An example of the dashboard is shown in Figure 5 where the user's activities during one day are presented. In particular, this figure shows the different sets of activities performed by the user to analyze the correlation between certain actions and their associated consumption (e.g., Meal Preparation will be linked to the consumption of the electric oven or the coffee machine). This representation of daily activities also helps to understand the normal routine of the monitored user and helps to identify any episodes of anomalous behaviour that deviate from usual patterns. Furthermore, by combining activity data with information about energy consumption, the users can easily understand how their behaviour impacts the energy performance of the buildings. This knowledge is also useful for an intelligent appliance scheduling algorithm that considers both the user's household activities and the calculated energy consumption patterns resulting from these routines. By doing so, it can provide personalized assistance that not only optimizes energy usage but also aligns with the user's lifestyle and preferences.

VI. Conclusions and Future Works

This paper proposes a DT framework with a user-centric perspective to address the personalized energy efficiency challenge in buildings. By integrating occupants' preferences and habits into the framework, awareness of their impact on building management and efficiency is enhanced. The comprehensive ontology developed in a Microsoft Azure environment enables the semantic representation of building and person entities and their relationships, laying the foundation for informed decision-making.

Future work will focus on expanding the ontology to encompass additional aspects of building operations for AAL systems, focusing on appliance scheduling and management. The system will be later fed with real-time data streams from real-building IoT devices. This will be expedient in testing the framework's dynamic adaptation to changing environmental conditions and user behaviours.

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